

SATHYANARAYANA RAMESH

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Skills

- Python | Django | Openstack | Java | JavaScript | C++ | C | MongoDB | MySQL | Node | jQuery | NoSQL | Perl | PHP |
- Bootstrap | Git | Azure | AWS | Jenkins | Cloud Computing | CI/CD | Unit Testing | Rabbitmq | Lambda | OOP | REST | API
- Microservices | Frontend | Backend | Full-Stack | React | English, Hindi, Kannada – All professional proficiency or above

Education

Master of Science • Major in Computer Science	<u>Western Michigan University</u> Kalamazoo, MI, USA	08/2023 - 04/2025 3.83 GPA
Bachelor of Engineering • Computer Science and Engineering	<u>Global Academy of Technology</u> Bengaluru, India	08/2015 - 07/2019 3.3 GPA

Experience

Research Assistant **Western Michigan University(Psychology Dept)** **MI, USA 01/2024 - 04/2024**

- Developing a web-based preference assessment tool catering to researchers and educators, aimed at evaluating the preferred rewards of autistic children. Utilized video trials to enable subjects to express their preferences independently, eliminating the reliance on verbal communication.
- The web tool was developed using HTML, React js, CSS3, and Bootstrap for website design to seamlessly adjust to mobile and tablet screens.

Software Engineer Lead- Professional I **Capgemini Engineering** **Bengaluru, India 03/2022 - 07/2023**

- Key contributor to the OpenStack Cinder driver for IBM storage area networks, continuously improving its functionality to meet the evolving demands of modern storage infrastructures.
- Expanded automation test coverage for new volume group functionality, enhancing the driver's reliability and stability, crucial for dev and test environments.
- Led a high-performing team of three members in the development of volume migration with flash copy, a pivotal feature for efficient data management.
- Managed hypervisors and created virtual machines in OpenStack, optimizing settings for performance and resource utilization. Developed automation scripts for efficient VM deployment
- Actively contributed to the open-source community, collaborating with peers, sharing insights, and marking IBM's presence in the open-source ecosystem.

Senior Systems Engineer **Infosys** **Bengaluru, India 08/2019 - 03/2022**

- Designed and implemented an API for response mail approvals, integrating seamlessly with Linux cron jobs for automated actions. Streamlined approval processes using Perl and SQL development skills.
- Created network templates for Cisco routers, focusing on both IOS and IOS XR platforms. These templates included configurations for Border Gateway Protocol (BGP), static routing, and Access Control Lists (ACLs), which were later populated with IP values and applied at the router level.
- Developed an API serving as a firewall for internal product APIs using Java and Spring Framework.
- Improved customer portals with user interface enhancements and developed supporting backend APIs using Flask.
- Developed a Python Django API for link utilization data, integrating RabbitMQ. Reduced data retrieval times using Redis caching, developing microservices and RESTful APIs.
- Utilized MongoDB for data storage with a Pymongo interface, ensuring code integrity through GitHub. Deployed applications seamlessly with CI/CD pipelines, using Jenkins for continuous deployment.

Software Engineer Intern **Infosys** **Mysore, India 01/2019 - 04/2019**

- Developed a full-fledged online food delivery system, featuring separate customer and restaurant interfaces.
- Implemented PHP for the server-side and MySQL as the choice of database.
- Proficiently managed user profiles, orders, menus, and secure payments for customers, and provided menu management and billing tools for restaurants.

Awards

- Recognized with “**Outstanding Contribution**” award for best contribution to the project in two consecutive quarters, underscoring a consistent commitment to excellence. (11/2022, 01/2023)
- Started as a fresher and received the “**Rookie of the Year**” award for rapid development of key projects. (07/2021)
- **Insta Award** was awarded for the best efforts in meeting a strict deadline for the development of Product API integration. (06/2021)

Certifications

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| • Introduction to Data Science – Coursera | (07/2022) | • Algorithmic Toolbox – Coursera | (06/2022) |
| • Cisco IOS XR – Udemy | (07/2022) | • Python Data Structures – Udemy | (03/2022) |

Projects

Advanced Lane Detection (*Digital Image Processing*)

- Developed a deep learning model to segment lanes from road images, enhancing detection accuracy.
- Integrated classical computer vision techniques, such as edge detection, to identify lane lines based on geometric properties.
- Combined these technologies to improve system reliability and efficiency in real-time lane detection.

RAG Chatbot (*Machine Learning*)

- Designed a custom chatbot system for analyzing text from PDF documents and images using LangChain, Gemini, and LLaMA models.
- Leveraged natural language processing techniques to ensure accurate and relevant responses to user queries.
- Implemented machine learning algorithms to enhance information retrieval and text analysis capabilities within the system.

Node and Network Monitoring Automation

- Automated node and network monitoring processes across multiple platforms, ensuring real-time notifications using a Python script with Paramiko.
- Developed a network monitoring script, setting execute permissions and configuring log file locations to track network activities effectively.
- Installed Terraform on Ubuntu and configured access for AWS and GCP to manage resources efficiently.

Others

- Volunteering for a research initiative within the Computer Science department focused on Immersive Experiential Learning to promote AI readiness across STEM disciplines. Contributing to the development of the 'AI in Daily Life' module. This involves writing and designing programs aimed at providing practical insights into the integration of AI technologies in everyday scenarios.